GCSE Physics (AQA)

Topic: Particle Model of Matter

SECTION 1 – Detailed Revision Notes

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1. The Particle Model

Key Idea

The **particle model** states that all substances are made of tiny particles (atoms or molecules) which are in constant motion.

States of Matter

State	Arrangement	Movement	Forces
Solid	Close together, regular	Vibrate about fixed positions	Strong
Liquid	Close together, irregular	Move past each other	Weaker than solids
Gas	Far apart, random	Move freely and rapidly	Very weak

Key Exam Phrase

The properties of solids, liquids and gases can be explained by the arrangement, movement and energy of particles.

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2. Density

Definition

Density is the mass per unit volume of a substance.

Equation

$$\textbf{Density} = \frac{\textbf{Mass}}{\textbf{Volume}}$$

$$\rho = \frac{m}{V}$$

Where:

- ρ = density (kg/m^3)
- m = mass (kg)
- V = volume (m^3)

Required Units

- Mass → kilograms (kg)
- Volume → cubic metres (m^3)
- Density → kg/m^3

Rearranged Equations

$$m = \rho V$$

$$V = \frac{m}{\rho}$$

Measuring Density

Regular solid:

1. Measure mass using a balance.
2. Measure dimensions.
3. Calculate volume.
4. Use density equation.

Irregular object (Required Practical):

1. Measure mass.
2. Submerge in measuring cylinder.
3. Volume displaced = object's volume.
4. Use density equation.

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3. Internal Energy

Definition

Internal energy is the total kinetic energy and potential energy of the particles in a system.

$$\textbf{Internal Energy} = \textbf{Kinetic Energy} + \textbf{Potential Energy}$$

Kinetic Energy

Energy due to movement of particles.

Potential Energy

Energy stored due to forces between particles.

Key Exam Phrase

Increasing temperature increases the average kinetic energy of the particles.

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4. Changes of State

Melting

Solid → Liquid

Freezing

Liquid → Solid

Boiling

Liquid → Gas (throughout liquid)

Evaporation

Liquid → Gas (surface only)

Condensation

Gas → Liquid

Sublimation

Solid → Gas (directly)

Energy and Changes of State

During a change of state:

- Temperature remains constant.
- Energy is transferred.
- Energy is used to break or form bonds between particles.

Key Exam Phrase

During a change of state, energy is transferred to change the potential energy of particles, not their kinetic energy.

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5. Specific Latent Heat

Definition

Specific latent heat is the energy required to change the state of 1 kg of a substance without changing its temperature.

Equation

$$E = mL$$

Where:

- E = energy (J)
- m = mass (kg)
- L = specific latent heat (J/kg)

Two types:

- Specific latent heat of fusion (melting/freezing)
- Specific latent heat of vaporisation (boiling/condensing)

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6. Specific Heat Capacity

Definition

Specific heat capacity is the energy required to raise the temperature of 1 kg of a substance by 1°C.

Equation

$$E = mc\Delta T$$

Where:

- E = energy (J)
- m = mass (kg)
- c = specific heat capacity (J/kg°C)
- ΔT = temperature change (°C)

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7. Gas Pressure

Gas pressure is caused by particles colliding with the walls of a container.

Key Idea

- Increasing temperature → particles move faster
- Faster particles → more frequent collisions
- More collisions → higher pressure

If volume is fixed:

[\text{Temperature increases → Pressure increases}]

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8. Diffusion

Definition

Diffusion is the spreading out of particles from an area of high concentration to low concentration.

Key Points

- Happens in gases and liquids
- Faster at higher temperatures
- Caused by random motion of particles

Key Exam Phrase

The rate of diffusion increases with temperature because particles have more kinetic energy.

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9. Brownian Motion

Brownian motion is the random movement of particles suspended in a fluid.

Importance

Provides evidence for the particle model.

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10. Required Practical – Density

Students must be able to:

- Measure mass using a balance.
- Measure volume of regular objects.
- Measure volume of irregular objects using water displacement.
- Calculate density.
- Rearrange the density equation.

Common GCSE 6-Mark Focus Areas

- Explaining changes of state using particle model.
- Explaining gas pressure in terms of particle collisions.
- Explaining diffusion using kinetic theory.
- Describing required practical methods clearly.

Essential Exam Vocabulary

- Kinetic energy
- Potential energy
- Internal energy
- Random motion
- Concentration gradient
- Specific latent heat
- Specific heat capacity
- Collisions
- Fixed volume
- Change of state

END OF SECTION 1